

# Synoptic CB: Porewater Nutrients

May 2026 Samples

2026-10-05

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# 1 Run Information

**Run date:** 08/26/2026,09/01/2026,09/18/2026

**Run by:** Delaina Meador, Melanie Giessner

**Samples processed:** May 2026

**Run notes:** Run on 09/02 ran out of rxn wells, DM unable to rerun the tray, SEAL said tray was completed; some values were still ELH.peChk conc. for VNOx was low compared to expected - 3rd party check flag:REASSESS, unsure why this happens. The CCV's for NH3 and PO4 were renamed on this run to 0.15 mg/L and 1 mg/L ammonia so that is changed in the code back to CCV. Some duplicates out of range, but still >60% within 10% of eachother, so we accept these data. There is one sample that I have that is not in the metadata: GCW\_TR\_202607\_LysB\_20cm It does not look like there are any other samples missing from the July data, so I think we just need to check the datasheet and see if someone forgot to mark it off.

**NOx files used:**

- SEAL\_COMPASS\_Synoptic\_NOx\_202605\_1.csv
- SEAL\_COMPASS\_Synoptic\_NOx\_202605\_2.csv
- SEAL\_COMPASS\_Synoptic\_NOx\_202605\_3\_202607\_1.csv

**NH3/PO4 files used:**

- SEAL\_COMPASS\_Synoptic\_NH3\_PO4\_202605\_1.csv
- SEAL\_COMPASS\_Synoptic\_NH3\_PO4\_202605\_2.csv
- SEAL\_COMPASS\_Synoptic\_NH3\_PO4\_202605\_3\_202607\_1.csv

## 2 Import Data & Clean

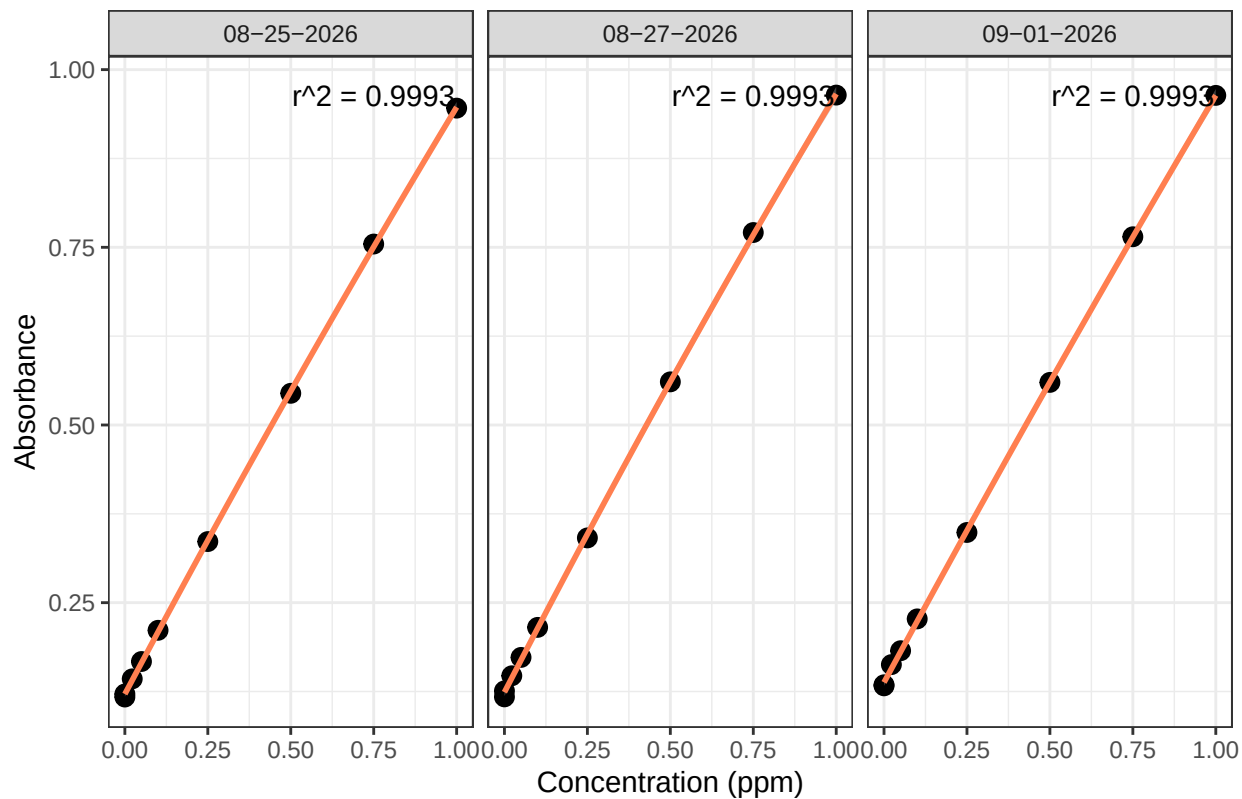
### 2.1 Fix any incorrectly entered sample data



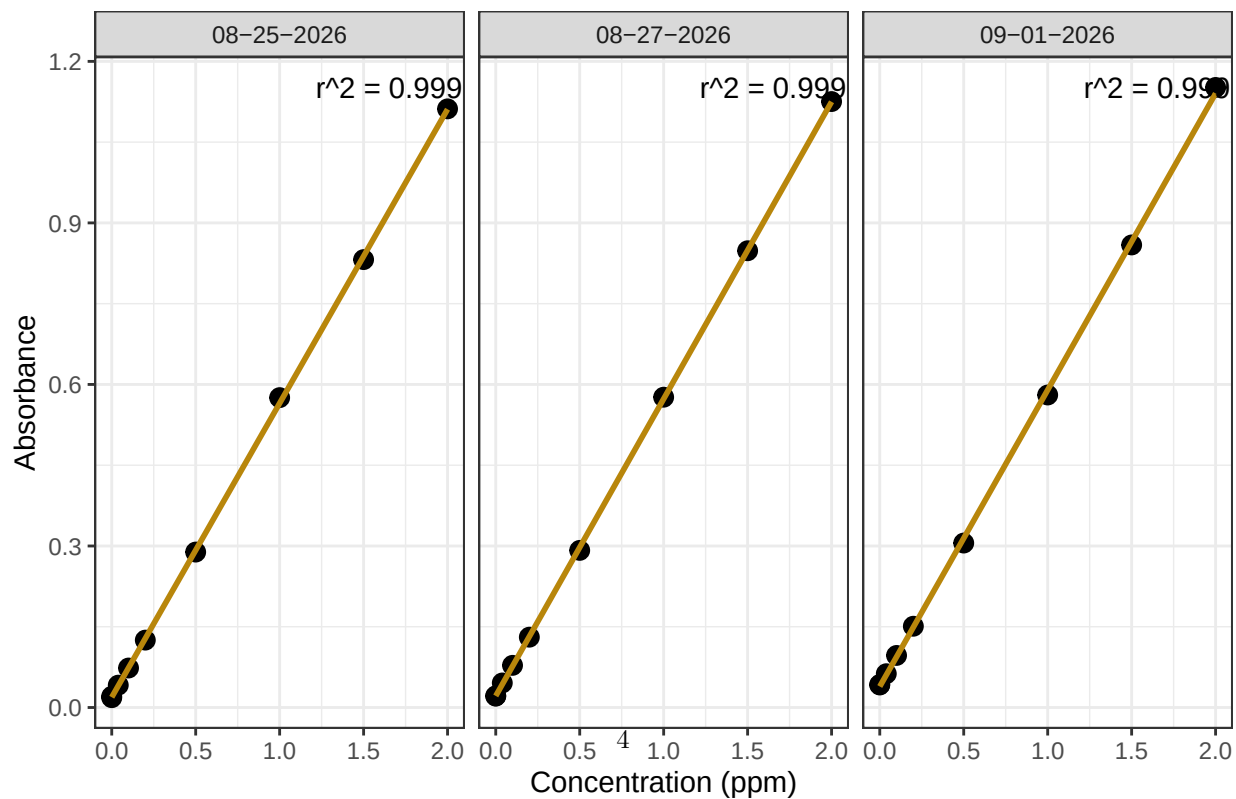
### 3 Run QAQC

#### 3.1 Assessing standard Curves

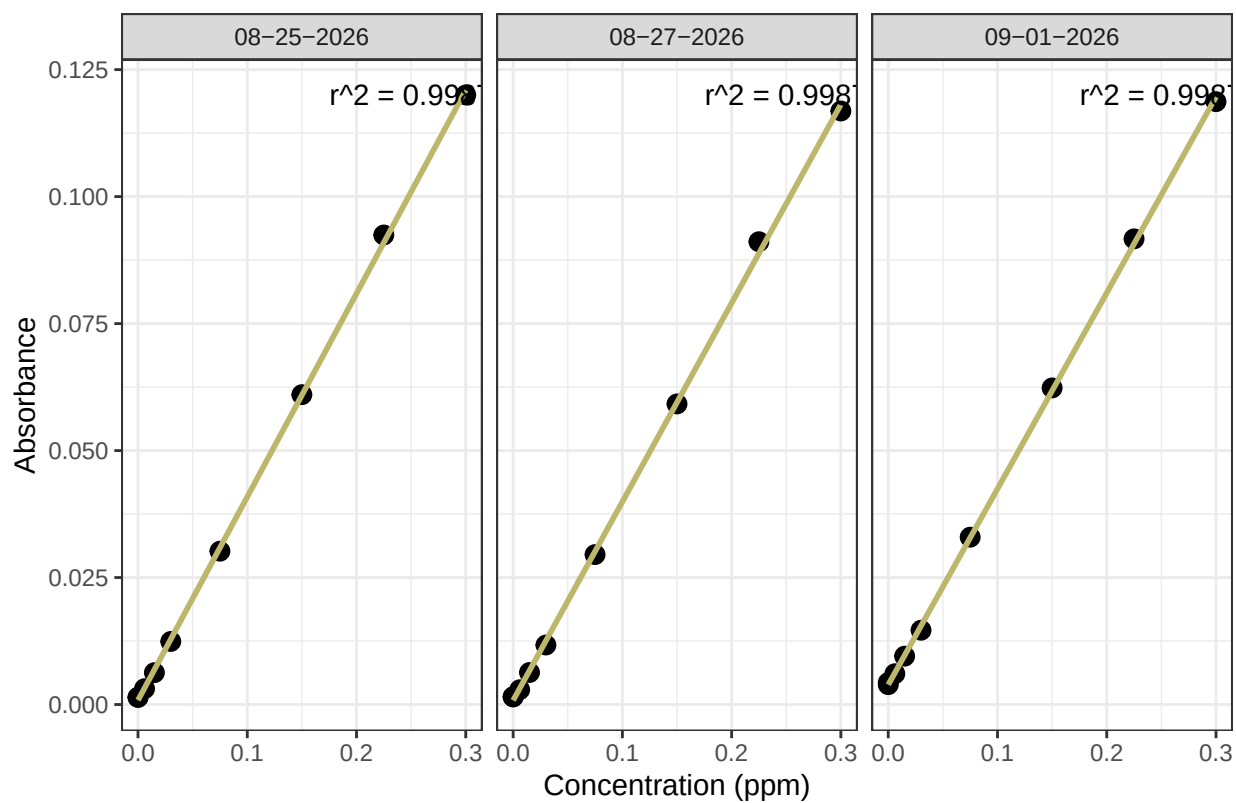
NOx Standard Curve



NH3 Standard Curve



PO4 Standard Curve

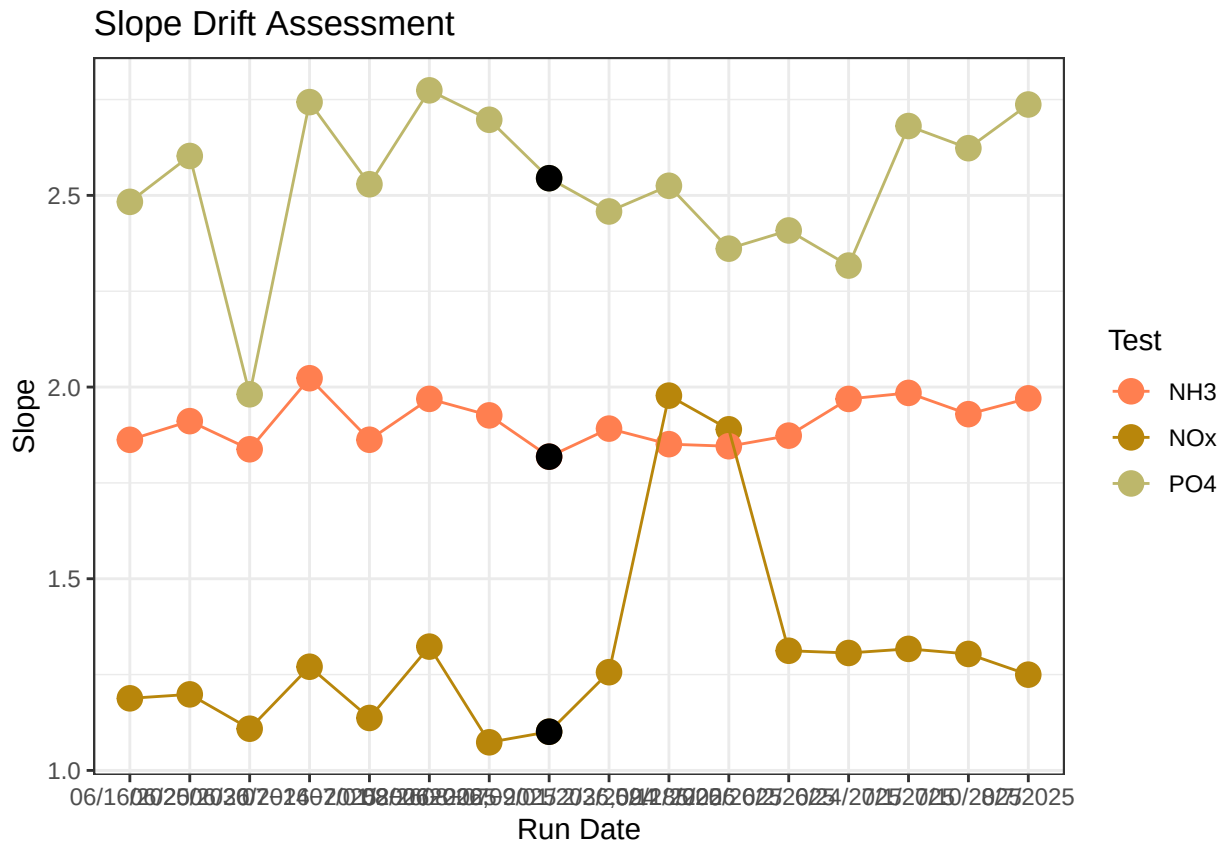


NOx Curve r2 GOOD - PROCEED

NH3 Curve r2 GOOD - PROCEED

PO4 Curve r2 GOOD - PROCEED

[1] "QAQC log file exists and has been read into the code."



| Test | Intercept | Slope | R_squared | Avg_Past_Slopes |
|------|-----------|-------|-----------|-----------------|
| NOx  | -0.142    | 1.101 | 0.999     | 1.313           |
| NH3  | -0.048    | 1.818 | 0.999     | 1.908           |
| PO4  | -0.005    | 2.545 | 0.999     | 2.529           |

### 3.2 Dilution Checks - If a sample was diluted, ensure we are using the diluted value

Duplicated samples:

| Sample_Name             |
|-------------------------|
| GCW_WC_202605_LysB_45cm |
| GCW_WC_202605_LysC_45cm |
| MSM_TR_202605_LysB_45cm |
| MSM_WC_202605_LysB_20cm |
| MSM_TR_202605_LysC_45cm |
| SWH_TR_202605_LysA_45cm |
| SWH_TR_202605_LysB_45cm |
| SWH_TR_202605_LysB_20cm |
| SWH_TR_202605_LysC_45cm |
| GCW_WC_202607_LysA_45cm |
| GCW_TR_202607_LysB_20cm |
| GCW_WC_202607_LysB_10cm |
| GCW_WC_202607_LysB_45cm |
| GCW_WC_202607_LysC_45cm |

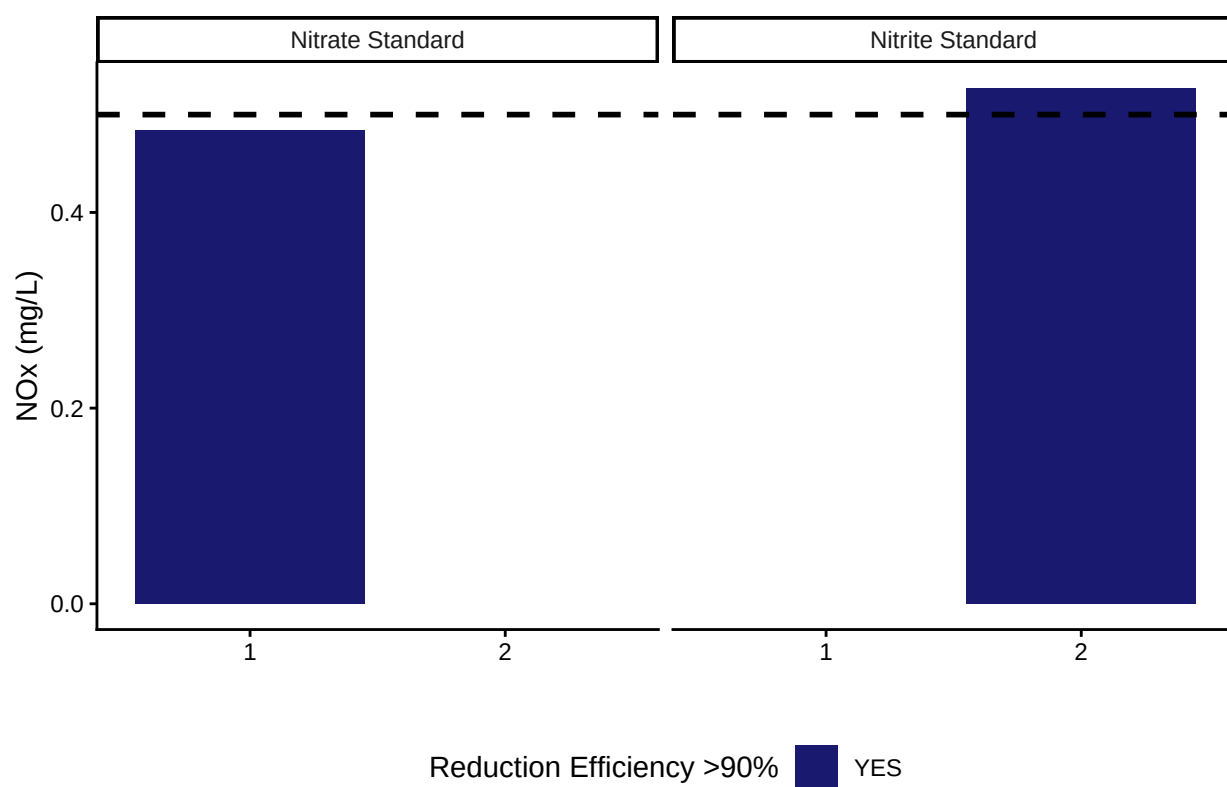
All duplicated samples have valid dilutions. No naming issues detected.

### 3.3 Performance Check

Table 1: Check Standard Results

| Analyte | Run mean (ppm) | Expected concentration (ppm) | Third Party Check flag       |
|---------|----------------|------------------------------|------------------------------|
| NOx     | 1.07           | 1.51                         | % Difference >25% - REASSESS |
| NH3     | 1.22           | 1.03                         | % Difference <25% - PROCEED  |
| PO4     | 0.90           | 0.82                         | % Difference <25% - PROCEED  |

### 3.4 Assess NOx Reduction Efficiency



Mean NOx Reduction Efficiency >95% - PROCEED

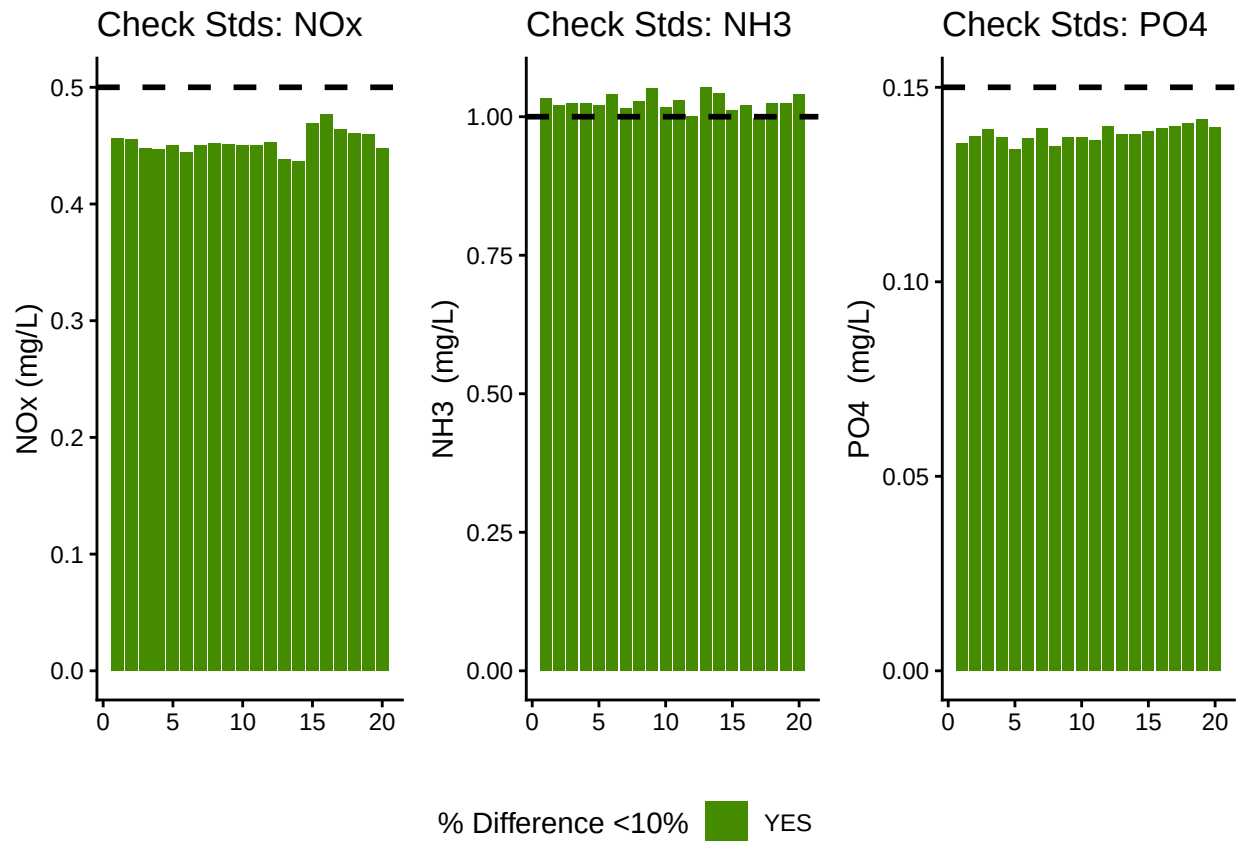
Mean Reduction Efficiency: 101.0903

### 3.5 Analyze the Check Standards

NOx Check Standard RSD within Range - PROCEED

NH3 Check Standard RSD within Range - PROCEED

PO4 Check Standard RSD within Range - PROCEED



>60% of NOx Check Standards are within range of expected concentration - PROCEED

>60% of NH3 Check Standards are within range of expected concentration - PROCEED

>60% of PO4 Check Standards are within range of expected concentration - PROCEED

### 3.6 Analyze Blanks

>60% of NOx Blanks are below the lower 25% quartile of samples or 1/2 detection limit - PROCEED

>60% of NH3 Blanks are below the lower 25% quartile of samples - PROCEED

>60% of PO4 Blanks are below the lower 25% quartile of samples- PROCEED

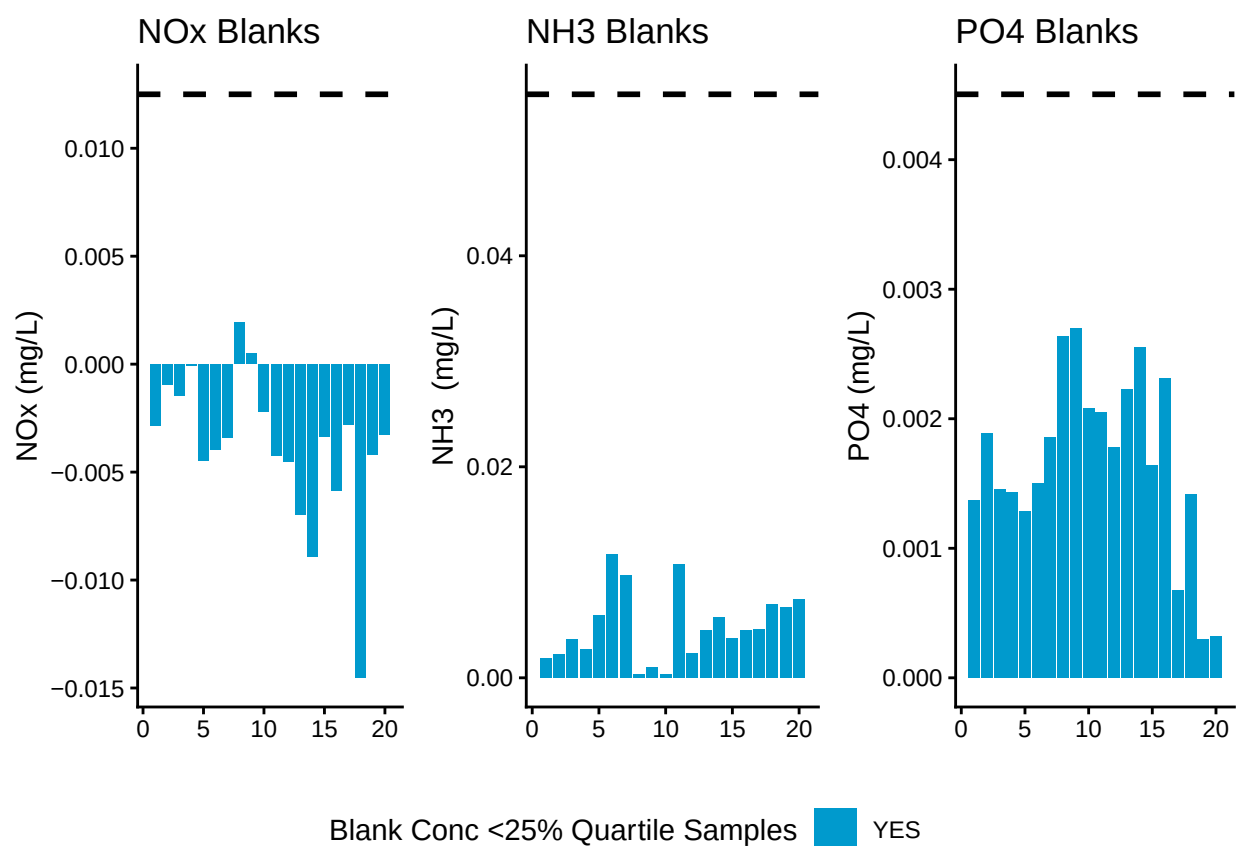


Table 2: Mean Concentration of Blanks

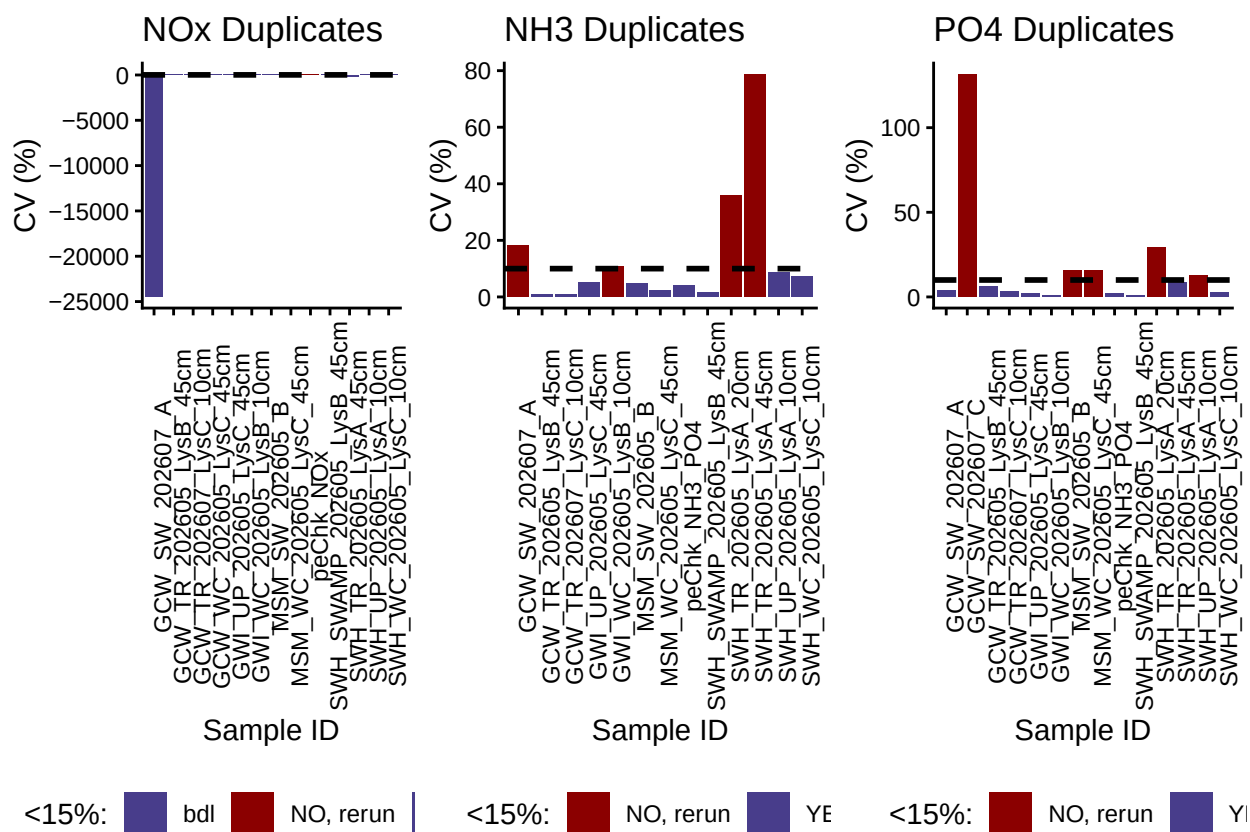
| Test | Blank_Mean_Conc |
|------|-----------------|
| NOx  | -0.0038         |
| NH3  | 0.0048          |
| PO4  | 0.0017          |

### 3.7 Analyze Duplicates

>60% of NOx Duplicates have a CV <10% - PROCEED

>60% of NH3 Duplicates have a CV <10% - PROCEED

>60% of PO4 Duplicates have a CV <10% - PROCEED

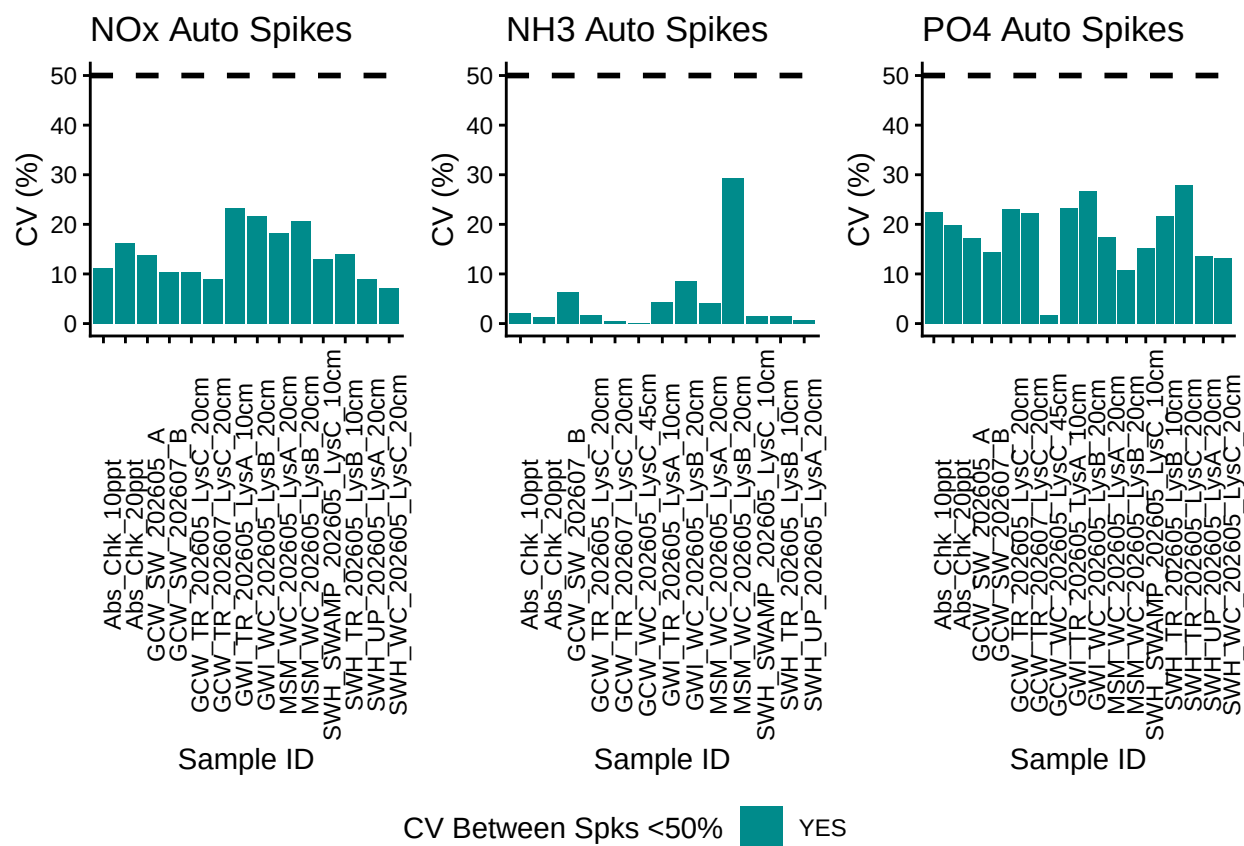


### 3.8 Spikes

>60% of Spikes have a CV <50% - PROCEED

>60% of Spikes have a CV <50% - PROCEED

>60% of Spikes have a CV <50% - PROCEED



### 3.9 Matrix Effects

Run either a 10ppt ASW or a 10ppt and a 20ppt ASW to assess matrix effects due to salinity

NO NOx Matrix Effect, PROCEED

NO NH3 Matrix Effect, PROCEED

NO PO4 Matrix Effect, PROCEED

### 3.10 Add unit converted cata column (convert mg/L to uM )

## 4 Concentration / Dilution Flags

4.1 Sample Concentration Flag - mark sample as within or out of range

4.2 Sample processing: pull out sample id

## 5 Metadata Matching

### 5.1 Check Sample IDs with Metadata / Sample log and merge metadata

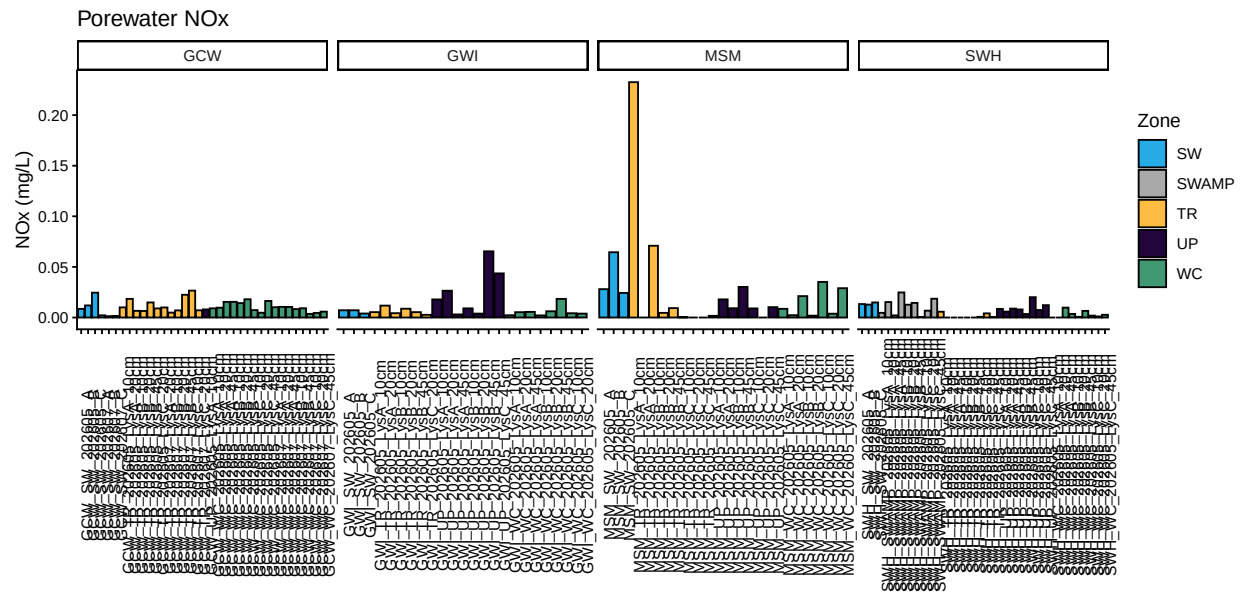
Some sample IDs are missing from metadata.

```
[1] "GCW_TR_202607_LysB_20cm"
```

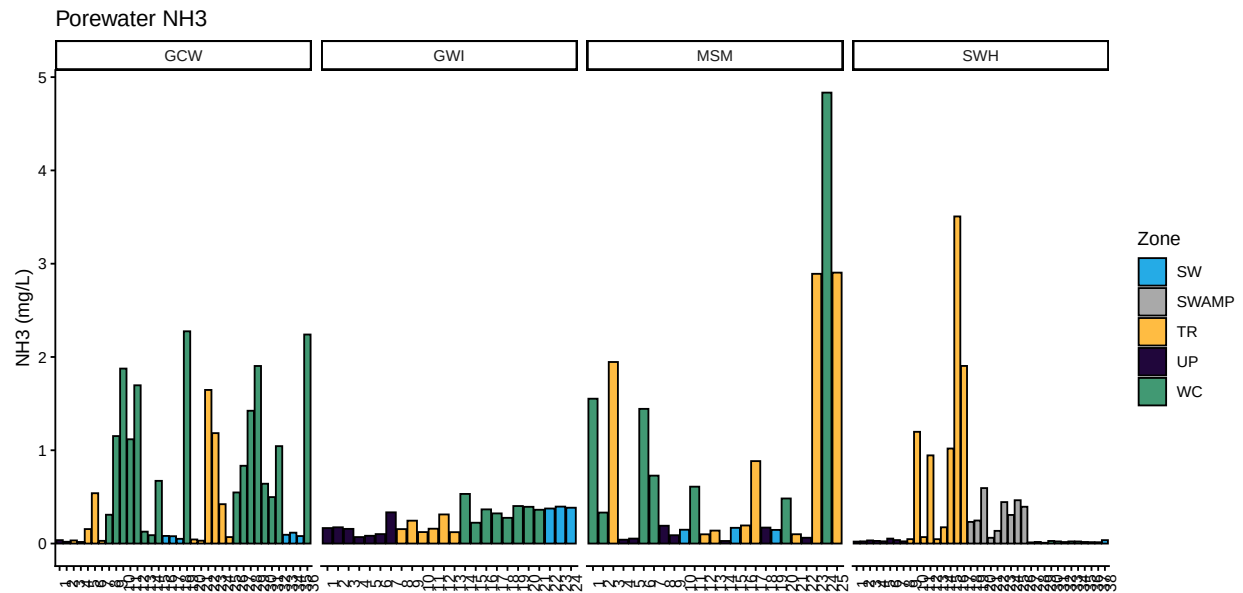
```
[1] "NUTR_ID"      "Year"         "Month"
[4] "Day"          "YearMonth"    "Site"
[7] "Zone"         "Lysimeter"    "LysID"
[10] "Depth_cm"     "Depth"        "Time (24hr)"
[13] "Time Zone_EDT/EST" "Field Notes"
```

## 6 Visualize Data

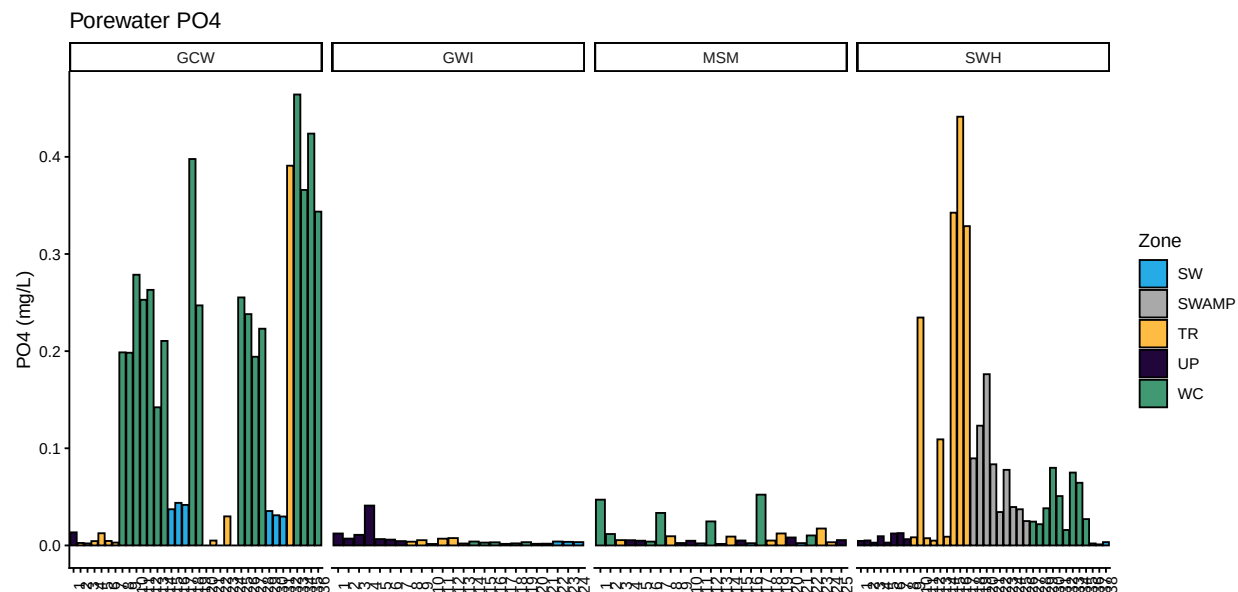
##NOx Samples



##NH3 Samples



##PO4 Samples



## 7 Export Processed Data